

Damage and recovery of the intrinsic and extrinsic foot muscles from running a full marathon

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Purpose: To investigate the effects of full marathon running on intrinsic and extrinsic foot muscle damage and to determine the relationship with the height change of the longitudinal foot arch following full marathon completion.

Methods: Magnetic resonance imaging-measured transverse relaxation time (T_2) of the abductor hallucis (ABH), flexor digitorum brevis (FDB) and quadratus plantae (QP), flexor digitorum longus (FDL), tibialis posterior (TP), and flexor hallucis longus (FHL) from 22 collegiate runners were assessed before and 1, 3, and 8 days after full marathon running. The three-dimensional foot posture of 10 of the 22 runners was further obtained using a foot scanner system before and 1, 3, and 8 days after the marathon.

Results: Marathon-induced increases in T_2 were observed in the QP, FDL, TP, and FHL 1 day after the marathon (+7.5%, +4.7%, +6.7%, and +5.9%, respectively), with the increased T_2 of TP persisting until 3 days after the marathon (+4.6%). T_2 changes of FDL and FHL from pre-marathon to DAY 1 showed direct correlations with the corresponding change in the arch height ratio ($r=0.823$, $p=0.003$, and $r=0.658$, $p=0.038$).

Conclusion: The damage and recovery response from a full marathon differed among muscles; QP, FDL, TP, and FHL increased T_2 after the marathon, whereas ABH and FDB did not. In addition, T_2 changes in FDL and FHL and the arch height ratio change were correlated. Our results suggest that the extrinsic foot muscles could be more susceptible to damage than the intrinsic during marathon running.

KEYWORDS

arch, foot, magnetic resonance image, marathon, running, T_2 relaxation time

1 | INTRODUCTION

Long-distance running is a popular exercise, and many runners participate in marathon races worldwide. Marathon

racing results in fatigue and induces decreased muscle strength of the lower extremities,¹ increased thigh and lower leg muscle hardness,²⁻⁴ and hamstring muscle damage with an intra-muscle difference at the muscle site.⁵

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The muscles of the foot can be grouped into intrinsic and extrinsic muscle groups depending on the origin of the muscle; the intrinsic muscles originate and insert within the foot and the extrinsic muscles originate in the lower leg and insert into the foot. These intrinsic and extrinsic foot muscles are thought to stabilize the medial longitudinal arch^{6,7} and cooperate with various body parts, including the plantar fascia.^{8,9} Previous studies have reported foot morphological changes, such as lowering of the medial longitudinal arch^{10,11} and more pronated posture of the foot caused by long-distance running.¹² Although it is challenging to associate morphological foot arch retention with intrinsic and extrinsic foot muscle functions, damage to intrinsic and extrinsic foot muscles, which is mostly identified by decreased muscle force output ability and muscle swelling/edema, has been suggested as a potential cause of foot morphological changes. However, the effect of marathon running on damage to intrinsic and extrinsic foot muscles remains to be proven. In the new concept of “core stability” of the foot, the intrinsic foot muscles work as primary and local stabilizers of the foot arches, whereas the extrinsic foot muscles work as global movers, and provide some stability to the arch.¹³ With this new concept, if intrinsic muscles play a primary role in supporting the foot arch, marathon-induced fatigue and/or damage to intrinsic foot muscles may be predicted in the lowered arch by prolonged running. However, we still lack information about the contribution and damage of intrinsic and extrinsic muscles following prolonged running. An exploration of this effect would be useful from both the standpoint of basic research and for practical runners.

The primary purpose of this study was to examine the effect of full marathon running on intrinsic and extrinsic foot muscles during recovery after a race. The secondary purpose was to examine the relationship between intrinsic and extrinsic foot muscle damage and arch height changes following a full marathon. We hypothesized that intrinsic and extrinsic foot muscles would be damaged by full marathon running and that this damage would be more prominent in the intrinsic foot muscle damages than in extrinsic muscles. This is because intrinsic foot muscles are assumed to support the foot longitudinal arch primarily, which would not be maintained after marathon running.

2 | MATERIALS AND METHODS

2.1 | Participants

This study was approved by the local institutional ethics committee and was conducted in accordance with the Declaration of Helsinki. All participants were informed of

the purpose, procedures, and risks of the study, and written informed consent was obtained from each participant. Twenty-three college runners (20 males and 3 females) participated in this study. The recruited participants belonged to a track and field club that participated in various recreational sports activities and/or running training two or three times weekly and had registered for a full marathon race (42.195 km, Mt. Fuji International Marathon in 2019 or 2021). The exclusion criteria included (a) a history of serious injuries or any operative treatment and (b) the presence of subjective symptoms that could interfere with sports activities.

2.2 | Study design

This study was planned as an observational study using a one-way analysis of variance. Assuming an effect size of $f=0.8$, $\alpha=0.05$, and $\beta=0.8$, the minimum number of participants required to establish a significant change in the T_2 relaxation time of intrinsic foot muscles was determined as 19 (G*power, version 3.1.9.2).

Participants visited the laboratory, and T_2 -weighted magnetic resonance (MR) image was obtained on two or four occasions: 3 days before (PRE), 1 day after (DAY 1), 3 days after (DAY 3), and 8 days after (DAY 8) the marathon race. Measurement was performed in all participants at the first two timepoints, while the latter two occasions were reserved for participants who could undergo follow-up ($n=16$). To avoid extra physical activity other than marathon running, participants were required to avoid running for training and in daily life, performing structured physical activity, or receiving special care for recovery, such as a massage, during the observation period. The marathon race was an official race of the Association of International Marathons and Distance Races and the Japan Association of Athletics Federations.

2.2.1 | MR image acquisition and analysis

Intrinsic and extrinsic T_2 MR imaging was performed on a 3.0-Tesla MR system using a 16-channel flex coil with knee and foot ankle positioner (Signa Premier; GE Healthcare) with the following parameters: field of view, 200×200 mm; matrix, 512×512 ; slice thickness, 5 mm; echo times, 25, 50, 75, and 100 ms; repetition time, 2500 ms. Participants were imaged in the supine position, with their right foot in the coil in a neutral position. All original digital images were rendered anonymous, and investigators were blinded to the measurement date. Image measurements were performed using open-source software (ImageJ; National Institute of Health). The abductor

hallucis (ABH), flexor digitorum brevis (FDB), and quadratus plantae (QP) were measured as the intrinsic foot muscles originating in an insert on the foot, and the flexor digitorum longus (FDL), tibialis posterior (TP), and flexor hallucis longus (FHL) were targeted as extrinsic foot muscles originating in the lower leg, crossing the ankle and inserting in the foot. Regions of interest for each of the individual foot (ABH, FDB, and QP) and lower leg (FDL, TP, and FHL) muscles were manually outlined. We selected the images at the level of the osseous insertion of the tibio-calcaneal ligament at the calcaneus, which was reported as best visualized for measurement of the cross-sectional area of the ABH, HDB, and QP (Figure 1A),¹⁴ and at the midpoint of the lower leg since there is a report that separately identified TP and FDL, which was not possible in more distal one-third of leg length (Figure 1B).¹⁵ T_2 relaxation time was calculated by least-squares analysis, fitting the signal intensity at each of the four echo times ($n \times 25$ ms: 25, 50, 75, and 100 ms) to a monoexponential decay using the following equation:

$$S_n = S_0 \exp(-TE_n/T_2) \quad (n = 25, 50, 75, \text{ and } 100 \text{ ms})$$

where TE is the echo time, S_0 is the signal intensity at 0 ms, and S_n is the signal intensity at TE_n .

2.3 | Toe flexor muscle strength

The maximum voluntary isometric toe flexor strength in the right foot was measured using a dynamometer for toe flexors (T.K.K 3365; Takei Scientific Instrument Co.) after MR scanning. The strength measured by this instrument has been reported to correlate with the maximal anatomical cross-sectional areas of the adductor hallucis oblique head, adductor hallucis transverse head, and FDB.¹⁶ Participants performed voluntary isometric contraction by gradually gripping the grip-bar of the dynamometer for 3 s with their toes in the sitting condition with their bare

foot approximately shoulder width apart and with 90° of hip and knee flexion and 0° ankle dorsi/plantar flexion. The measurements were repeated three times, and the average value was used for further analysis.

2.4 | Longitudinal foot arch height

The right foot posture of 10 participants were scanned using a three-dimensional foot scanner (JMS-2100CU; Dream GP) at PRE, DAY 1, DAY 3, and DAY 8 following the previously reported procedure.^{10,11} These 10 runners were participants in 2019 out of the 23 participants who participated in this study in 2019 and 2021. The most medial point of the navicular bone was identified by palpation and marked prior to scanning. The participants were asked to maintain a relaxed upright position with their feet bare, approximately shoulder width apart, and were asked to place their non-scanning foot on a platform, which was adjusted at the same height as the foot bearing area of the foot scanner so that the weight was evenly distributed on both sides during scanning. The foot length, the length between the midpoint of the heel and the head of the first or second digit (whichever was longer), and the navicular bone mark height from the floor were automatically recognized and calculated by the foot scanner system. The arch height ratio was calculated as the navicular bone height divided by foot length by 100 (%).

2.5 | Statistical analysis

Statistical analysis was performed using SPSS version 25 (IBM). The results are reported as means and standard deviations (SD), unless stated otherwise. The Friedman test was performed to determine whether significant differences exist in T_2 relaxation time and arch height ratio between the measurement points. This non-parametric test was applied as the Shapiro–Wilk test revealed some

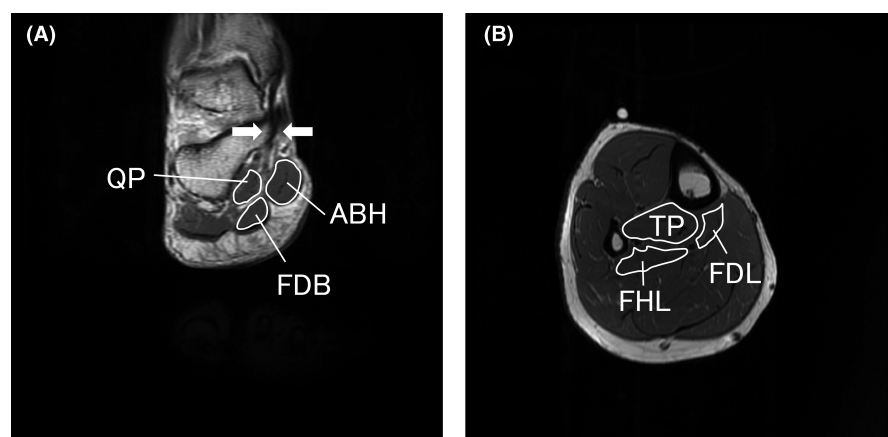


FIGURE 1 Example T_2 magnetic resonance images of the foot (A) and leg (B). Arrowheads in (A) show an osseous insertion of the tibio-calcaneal ligament. ABH, abductor hallucis; FDB, flexor digitorum brevis; FDL, flexor digitorum longus; FHL, flexor hallucis longus; QP, quadratus plantae; TP, tibialis posterior.

data were not normally distributed. When significance was detected, post-hoc tests (Wilcoxon signed-rank tests with Bonferroni adjustment) were performed. Spearman's rank order correlations were calculated to examine relationships between changes in the T_2 relaxation time of intrinsic and extrinsic foot muscles and changes in the arch height ratio on DAY 1 from PRE. Statistical significance was set at $p < 0.05$.

3 | RESULTS

Twenty-two participants (19 males and 3 females) completed the full marathon, and data from one participant who dropped out of the race were excluded. Participants' demographic data are shown in Table 1.

There were significant differences in T_2 relaxation time between timepoints in the QP ($X^2(3)=11.941$, $p=0.008$), FDL ($X^2(3)=9.322$, $p=0.025$), TP ($X^2(3)=22.146$, $p<0.001$), and FHL ($X^2(3)=12.448$, $p=0.006$), but not in the ABH ($X^2(3)=7.452$, $p=0.590$) or FDB ($X^2(3)=4.929$, $p=0.177$) throughout the observation period. The post hoc test showed that the T_2 relaxation time of the QP, FDL, TP, and FHL significantly increased from PRE to DAY 1 after the marathon (Figure 2). The absolute values of T_2 relaxation times of the intrinsic foot muscles were ABH: 28.21 ± 1.53 ms (PRE), 29.18 ± 2.80 ms (DAY 1), 27.75 ± 2.10 ms (DAY 3), and 28.10 ± 1.95 ms (DAY8); FDB: 28.28 ± 1.43 ms (PRE), 28.99 ± 2.72 ms (DAY 1), 27.85 ± 1.82 ms (DAY 3), and 27.28 ± 2.64 ms (DAY8); QP: 29.17 ± 1.29 ms (PRE), 31.22 ± 3.07 ms (DAY 1), 29.60 ± 2.51 ms (DAY 3), 29.21 ± 2.18 ms (DAY8), respectively. The absolute values of T_2 relaxation times of the extrinsic foot muscles were FDL: 27.81 ± 1.28 ms (PRE), 29.07 ± 1.30 ms (DAY 1), 28.29 ± 1.52 ms (DAY 3), and 27.58 ± 1.38 ms (DAY 8); TP: 28.11 ± 1.26 ms (PRE), 29.94 ± 1.38 ms (DAY 1), 29.60 ± 1.22 ms (DAY 3), and 28.67 ± 1.38 ms (DAY8); FHL: 27.41 ± 1.39 ms (PRE), 28.95 ± 1.60 ms (DAY 1),

28.31 ± 1.18 ms (DAY 3), and 27.71 ± 1.65 ms (DAY 8), respectively.

No significant difference was observed from PRE to DAY 8 in toe flexor strength ($X^2(3)=4.401$, $p=0.221$) (Figure 3).

The arch height ratio had statistical significance ($X^2(3)=13.848$, $p=0.003$) and it decreased from PRE to DAY 1 and DAY 3 (Figure 4). Table 2 presents the Spearman's correlations between full marathon-induced changes in T_2 relaxation time of intrinsic and extrinsic foot muscles and changes in the arch height ratio on DAY 1 from PRE. Positive correlations were demonstrated between the changes in the arch height ratio and T_2 relaxation time change in FDL and FHL.

4 | DISCUSSION

The main findings of this study were that (1) only the intrinsic QP muscle and the extrinsic FDL, TP, and FHL foot muscles were damaged by full marathon running, and (2) T_2 relaxation time changes of FDL and FHL from PRE to DAY 1 positively correlated with changes in the arch height ratio from PRE to DAY 1. Therefore, although our first hypothesis was partially supported, the second hypothesis was not supported by the results. These results suggest that the damage induced by prolonged running was less prominent in the intrinsic foot muscles than in the extrinsic muscles.

To the best of our knowledge, this is the first study to report the T_2 values of intrinsic foot muscles; therefore, our T_2 values of intrinsic foot muscles and those of prior studies cannot be directly compared. The average T_2 values of extrinsic foot muscles in this study were generally lower compared to previous studies; FDL/FHL as a whole: 35.8 ms and TP: 34.56–37.76 ms before ankle plantar exercise,^{17–19} although we could not determine what this baseline discrepancy reflects.

The ABH, FDB, and QP are the three largest plantar intrinsic foot muscles that regulate foot longitudinal arch stiffness in response to the magnitude of the vertical ground reaction force.²⁰ Different activation patterns of these three muscles while running have been previously reported: the ABH is activated from late swing to early stance; subsequently, the FDB and QP activate at foot contact, peaking at mid-stance and continuing into the propulsion phase.²⁰ A possible explanation for the more prominent damage in the extrinsic foot muscles could be due to the potential interpretations as follows. The more prominent damage to extrinsic foot muscles may reflect the relatively larger work on the ankle joint. Previous studies reported that the intrinsic foot muscles were involved in the push-off of the ground for locomotion^{21,22}

TABLE 1 Demographic data of the participants for T_2 MR imaging ($n=22$) and foot posture scanning ($n=10$).

Variable	(n = 22)		(n = 10)	
	Mean	SD	Mean	SD
Age (years)	21.4	1.9	21.9	2.3
Height (cm)	170.4	9.0	171.3	6.9
Weight (kg)	60.1	7.5	61.9	7.5
BMI (kg/m ²)	20.7	1.9	21.1	2.3
Marathon time	4 h 09 min 44 s	1 h 02 min 43 s	4 h 35 min 50 s	54 min 41 s

Abbreviations: BMI, body mass index; MR, magnetic resonance; SD, standard deviation.

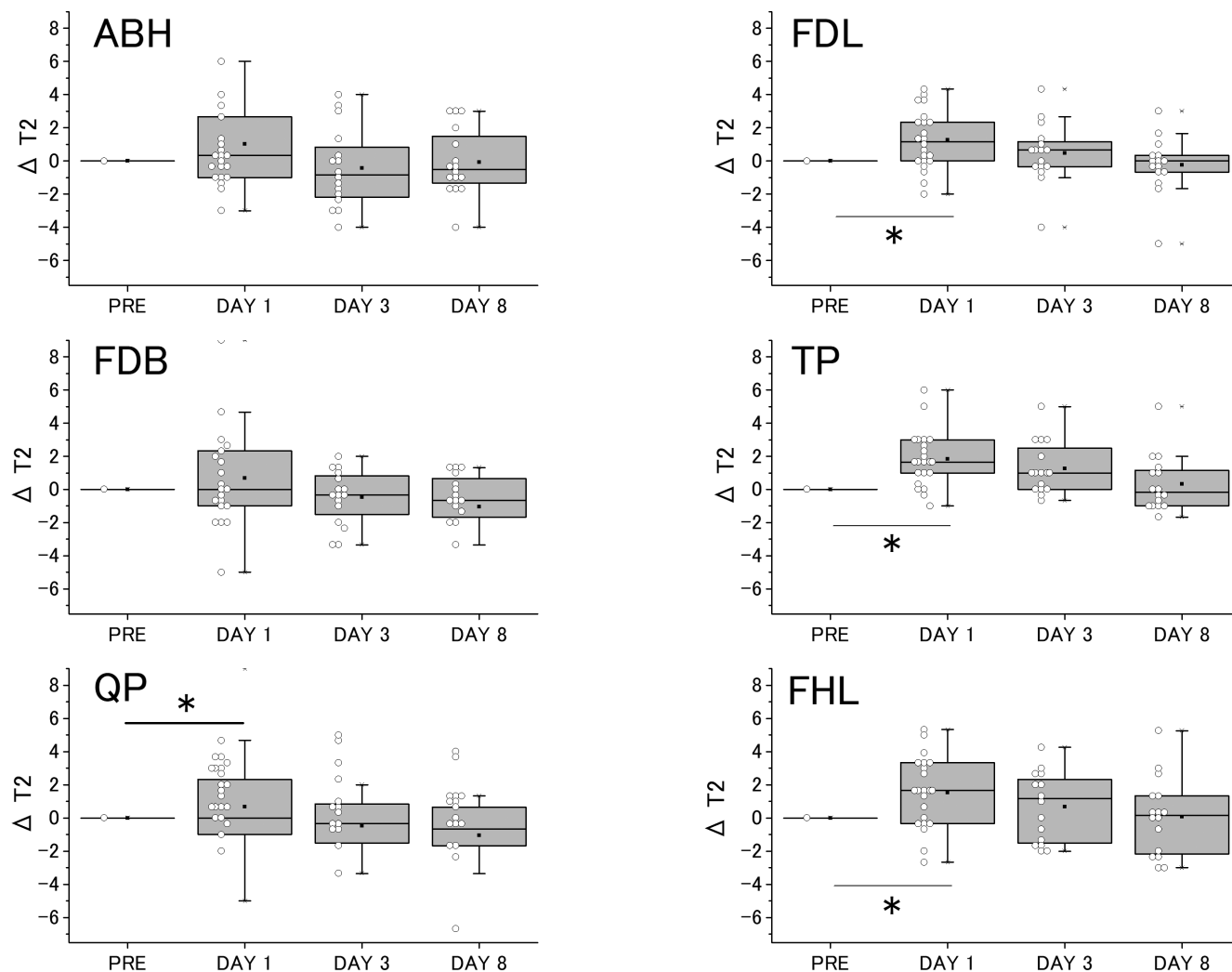


FIGURE 2 Changes in T_2 relaxation time of each intrinsic and extrinsic muscle at 3 days before (PRE), 1 day after (DAY 1), 3 days after (DAY 3), and 8 days after (DAY 8) the marathon race. Data are plotted as the individual raw change (Δ) value from PRE (white circle), with a box-whisker plot with median (black bar) and mean (black square) and its 25th and 75th percentiles. The whiskers show extreme values. *significant difference (vs. PRE; $p < 0.05$); $n = 22$ for PRE and DAY 1 and $n = 16$ for DAY 3 and DAY 8. ABH, abductor hallucis; FDB, flexor digitorum brevis; FDL, flexor digitorum longus; FHL, flexor hallucis longus; QP, quadratus plantae; TP, tibialis posterior.

and the positive foot work by plantar flexor muscles increased throughout long running²³ whereas FDL, FHL, and TP were recruited to ankle plantar flexion exercise along with the gastrocnemius.¹⁷ However, upon comparing the ankle and foot positive work values reported in a study by Honert et al.,²³ the positive ankle work was found to be almost two times that of the foot. Additionally, the reported normative activation pattern of foot intrinsic muscles was measured using a treadmill,²⁰ and the patterns of muscle activity may differ in overground marathon running, although there appears to be no research on comparing activity patterns of intrinsic foot muscles between treadmill and overground running. Moreover, the toe flexion strength in this study was comparable across the four measurement occasions. This was unexpected, since previous studies reported that the muscle strength

of knee flexion,⁵ knee extension, and plantar flexion¹ muscles decreased after marathon running, which was inferred as marathon-induced fatigue. Collectively, our results indicate that the intrinsic ABH and FDB foot muscles may not become fatigued after a full marathon due to the lack of a significant difference in T_2 relaxation time and toe flexion torques on DAY 1, 3, and 8 from PRE.

The QP muscle has a similar anatomical pathway to the medial and central slips of the plantar aponeurosis but has several unique anatomical characteristics. A recent cadaveric study found that the QP attaches to the FDL and/or FHL, indicating that the QP functions as a counter to the oblique pull of the FDL and FHL and as a plantar flexor of the hallucis and the lesser digits.²⁴ In this study, the QP was the only intrinsic muscle that showed a significant increase in T_2 relaxation time, while significant increases

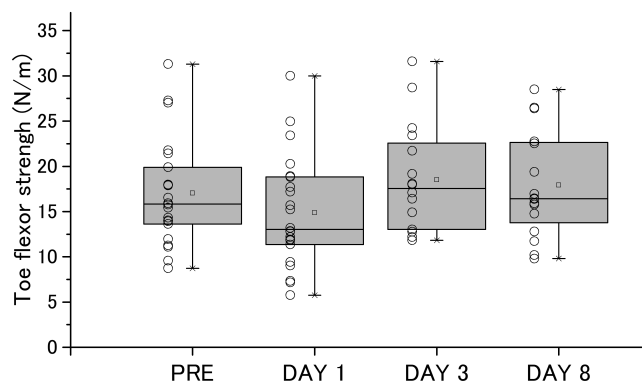


FIGURE 3 Changes in toe flexor strength 3 days before (PRE) and 1 day after (DAY 1), 3 days after (DAY 3), and 8 days after (DAY 8) the marathon race. Data are plotted as individual values (white circle), with a box-whisker plot with median (black bar) and mean (black square) and its 25th and 75th percentiles. The whiskers show extreme values.

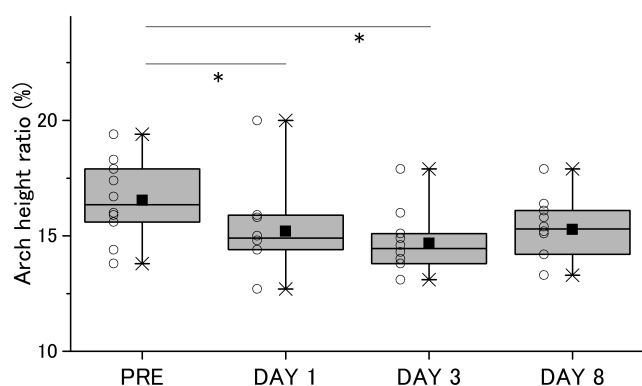


FIGURE 4 Change in the arch height ratio 3 days before (PRE) and 1 day after (DAY 1), 3 days after (DAY 3), and 8 days after (DAY 8) the marathon race. Data are plotted as individual values (white circle), with a box-whisker plot with median (black bar) and mean (black square) and its 25th and 75th percentiles. The whiskers show extreme values. *significant difference (vs. PRE; $p < 0.05$).

TABLE 2 Spearman's correlations between changes in T_2 of the intrinsic and extrinsic foot muscles and changes in the arch height ratio on 1 day after the marathon from 3 days before.

	Correlation	Significance
ABF	0.340	0.340
FDB	0.043	0.907
QP	-0.177	0.625
FDL	0.823	0.003*
TP	0.404	0.247
FHL	0.658	0.038*

Abbreviations: ABH, abductor hallucis; FDB, flexor digitorum brevis; FDL, flexor digitorum longus; FHL, flexor hallucis longus; QP, quadratus plantae; TP, tibialis posterior.

* $p < 0.05$.

were observed in all measured extrinsic muscles, including the FDL, TP, and FHL. This suggests that the QP may have an auxiliary function in conjunction with extrinsic foot muscles while running.

Furthermore, we identified correlations between changes in T_2 relaxation time in FDL and FHL and changes in the arch height ratio before and after full marathon running; feet with greater drops in navicular height had greater T_2 increases in the FDL and FHL after full marathon running. The significant decrease in arch height ratio aligned with previous studies,^{10,11} and its mean change of 1.8 in this study was in the range of 1.2–2.7 that comes from previous studies using the same measurement instrumentation and procedure.^{10,11} The correlation of this study indicates that the damage to the FDL and FHL could be a factor that lower the foot longitudinal arch following marathon running. However, the asynchronous nature of the duration of the increased T_2 relaxation time of the QP, FDL, and FHL and the duration of decreased arch height ratio that remained for 3 days in this study and the previously reported duration for >8 days¹⁰ after a full marathon should be considered. Furthermore, Shiotani et al.²⁵ reported that plantar fascia, which is a stabilizer of the foot arch, stiffened after running 10 km. Consequently, further studies are required to examine whether the QP, FDL, and FHL muscle function in arch stabilization and to define the synergetic action of these muscles with other arch stabilizers during running.

In this study, all measured FDL, TP, and FHL muscles were damaged by full marathon running, indicating that extrinsic foot muscles are vulnerable to running. This result aligns with a previous study using ultrasonic shear-wave elastography, which showed that muscle stiffness of the FDL and TP increased after 30 min of running.²⁶ Previous studies have reported normal extrinsic foot muscle activity and its function during gait. The TP has been shown to have pulsatile activation during heel contact and in mid-stance,²⁷ and isometric contraction of FHL and FDL to maintain the longitudinal arch during the stance phase of gait has been reported using cadavers with a robotic gait-simulating apparatus.²⁸ The muscle activities of the TP and FDL during running in footwear were analyzed using fine-wire electromyography and showed no peak amplitude in the contact phase and relatively higher muscle activities in the propulsion phase than in the contact phase of running,²⁹ although the previous study did not perform a statistical comparison between the contact and propulsion phases of running. Anbarian et al. reported that the time allocation of the stance phase of running for the contact phase was reduced while increasing the forefoot push-off phase by a running-induced fatigue state.³⁰ Therefore, the demand

for muscle activity of these muscles may increase as running mileage increases. Since the increased stiffness of the FDL and TP was proposed as a factor driving medial tibial stress syndrome,³¹ recovery from marathon-induced fatigue of extrinsic muscles may be important to prevent running-related injuries.

This study has several limitations that should be noted. First, our study participants were relatively homogeneous of healthy young runners. Therefore, our results cannot be generalized to different age groups or injured runners, as their muscle morphology and strength and/or running kinematics would be different. Second, this study focused on intrinsic and extrinsic foot muscles; however, the human foot is a complex structure consisting of numerous bones, muscles, tendons and ligaments, and a plantar heel pad, and there are numerous potential contributing factors to marathon-induced fatigue of the lower extremity muscles and/or lowering longitudinal foot arch, including running speed, running pattern, and footwear. Therefore, further studies should include other components of the foot in their investigations and should examine the effects of other possible contributing factors to better understand the effect of full marathon running. Third, the participants wore their footwear to prevent unexpected injuries due to sudden footwear changes; however, the choice of footwear can affect running mechanics and economy.³² Therefore, future studies should consider the involvement of footwear with different types and/or cushion properties.

In summary, the damage and recovery response from a full marathon differed between muscles. All the measured extrinsic foot muscles, including the FDL, TP, and FHL, were damaged by running a full marathon. Of the intrinsic foot muscles, the QP, but not the ABH and FDB, were damaged by running. In addition, T_2 increases in FDL and FHL and the arch height ratio change were correlated. These results suggest that marathon-induced damage was more prominent in the extrinsic than the intrinsic foot muscles and could be associated with the lowering of the foot longitudinal arch.

5 | PERSPECTIVE

This study demonstrated that marathon-induced damages appear as T_2 increase in the FDL, TP, FHL, and QP, and can provide the basis for developing better recovery schemes focusing on these muscles before resuming daily training for runner's conditioning and/or injury prevention. Moreover, the correlations between T_2 increases in FDL and FHL and the arch height ratio change were observed. This finding might indicate a potential association between FDL and FHL and structural foot arch retention.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest associated with this manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

FUNDING STATEMENT

N/A.

PATIENT CONSENT STATEMENT

All participants were informed of the purpose, procedures, and risks of the study, and written informed consent was obtained from each participant.

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